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Problem Set: #1

Due Date: 02/03/2026

Acknowledgement: No help

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.

Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

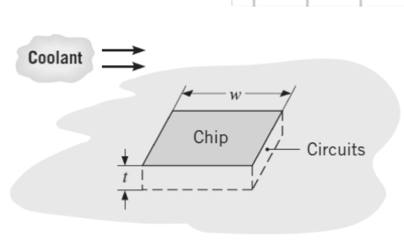
case 1:

Outside the atmosphere, we can presume no gases or fluids exist (vacuum). In this case, the radiation from surrounding bodies like the sun is the primary form of heat transfer.

case 2:

The dominant heat transfer mode is convection. Convection can be described as heat transfer that occurs between a surface & a fluid that is moving. In this case, the surface is moving & the atmospheric gases (fluids) are heavily perturbed by the motion of the spacecraft, creating heat transfer via convection.

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.



(a) What is the steady-state temperature difference between back and front surfaces of the chip?

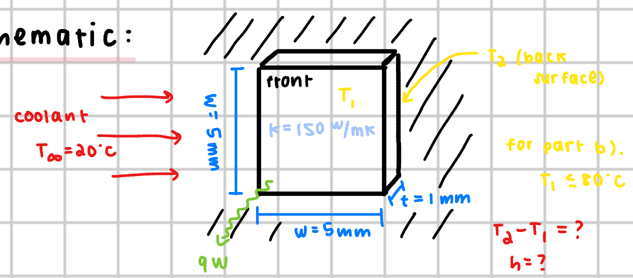
(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

Known: width of chip, thickness of chip, amount of energy dissipated under steady-state operation, coolant temperature, thermal conductivity of chip

Find: a). steady-state temperature difference between front & back of chip

b). if front surface temperature cannot be above 80°C, find minimum heat transfer coefficient & compare to textbook to determine convection processes that can be used to cool this chip

Schematic:



Assumptions: steady-state, constant coolant flow, negligible thermal radiation, uniform chip surface properties (other than side & back being insulated)

Analysis:

a). we know that the heat rate equation based on Fourier's law is

$$q = -kA\frac{dT}{dx}$$

we can write this out as

$$q = kA\frac{\Delta T}{t}$$

rearranged as

$$\Delta T = \frac{qt}{kA}$$

plugging in values

$$T_2 - T_1 = \frac{9W(0.001m)}{150\ W/(m\cdot K)\cdot 25\times10^{-6}\ m^2}$$

$$= 2.4K$$

b). we know that the convection heat rate equation is

$$q = hA(T_s - T_\infty)$$

rearranging

$$h = \frac{q}{A(T_s - T_\infty)}$$

plugging in values

$$h = \frac{9W}{25\times10^{-6}m^2(80°C - 20°C)}$$

$$= 6000\ W/m^2K$$

Referencing table 1.1, we can see this value for h corresponds to the range of forced convection with liquids & convection with phase change by boiling or condensation. These would both be valid processes by which convection can be used to cool the chip.

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha = 0.76$.

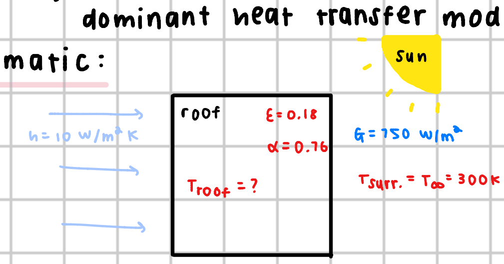
(a) Consider a day of solar irradiation $G = 750$ W/m^2, ambient temperature $T_\infty = 300$ K, surrounding temperature $T_{surr} = 300$ K, and heat transfer coefficient of air flow $h = 10$ W/(m$^2\cdot$ K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

Known: $\varepsilon = 0.18$, $\alpha = 0.76$, $G = 750$ W/m^2, $T_\infty = 300$K, $T_{surr} = 300$K, $h = 10$ W/m^{2}K

Find: a). temperature of the roof, is the cat comfortable walking on this roof
b). dominant heat transfer mode from a).

Schematic:



Assumptions: steady-state, bottom surface of roof is thermally insulated, uniform heating & heat transfer across roof

Analysis:

a). we will do an energy balance here

$$\dot{q}_{in} = \dot{q}_{out}$$

$$\underset{\text{radiation absorbed}}{\alpha G} = \underset{\text{convection}}{h(T_s - T_\infty)} + \underset{\text{radiation emitted}}{\varepsilon\sigma(T_s^4 - T_{surr}^4)}$$

$$\alpha G = hT_s - hT_\infty + \varepsilon\sigma T_s^4 - \varepsilon\sigma T_{surr}^4$$

$$hT_s + \varepsilon\sigma T_s^4 = \alpha G + hT_\infty + \varepsilon\sigma T_{surr}^4$$

we can plug in values & use a graphing calculator to solve

$$10T_s + 0.18(5.67\times10^{-8})T_s^4 = 0.76(750) + 10(300\text{K}) + 0.18(5.67\times10^{-8})(300\text{K})^4$$

$$10T_s + 1.0206\times10^{-8}T_s^4 = 3652.6686$$

$$T_s = 349.96\text{K}$$

No, this far too hot for a cat. According to studies, the temperature that a cat can at most withstand is 52°C. this temperature we calculated is much too high.

b). Let's compare the amount of radiation absorbed to convection & radiation emitted.

we know $T_s = 349.96$K

looking at the 2 temperature dependent terms in the balance above

$$h(T_s - T_\infty) = 499.6 \text{ W/m}^2$$

$$\varepsilon\sigma(T_s^4 - T_{surr}^4) = 70.42 \text{ W/m}^2$$

we can see the convection term is significantly larger, making it the dominant heat transfer mode.